

Sharon Mlotshwa | AI Researcher

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With over 4 years of research experience, I've worked across pediatric radiology and ophthalmology research projects, co-authoring peer-reviewed work in MICCAI, WACV and other journals. My journey has intersected clinical research, medical imaging, and logistics, across groundbreaking research institutions like MBZUAI in Abu Dhabi, vibrant healthcare research labs like Real Doctor AI Research Center in China and global trade and logistic companies like Abu Dhabi Ports Group. I bring technical depth, a researcher's rigour, a collaborator's qualities and a genuine passion to contributing to impactful PhD level research.

SKILLS

Research Projects

Diabetic Retinopathy Grading, Bone Age Assessments, Medical Computer Vision, Model Generalization

Technical Skills

Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Scikit-learn, PyTorch, Git, HPC, Databases, $\LaTeX$

Data & Analysis

Exploratory Data Analysis, Data Cleaning, Data Visualization

Soft Skills

Communication, Ownership, Collaboration, Adaptability, Problem-solving, Coachable, Scientific Writing

EDUCATION

MSc in Machine Learning | CGPA - 3.82/4.0

Mohamed bin Zayed University of Artificial Intelligence, UAE

Aug 2022 — Jun 2024

Advisor - Dr. Muhammad Haris Khan

BEng in Computer Science and Technology | CGPA - 4.44/5.0

Zhejiang University of Technology, China

Sep 2015 — Jun 2019

Advisor - Dr. Pengyi Hao

EXPERIENCE

Clinical Research – Domain Generalizable Diabetic Retinopathy Grading (MSc Research)

Aug 2022 — May 2024

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

- Conducted healthcare research focused on Domain Generalization in Diabetic Retinopathy Grading, enhancing robustness of AI models across diverse patient populations and medical imaging conditions.
- Co-authored 3 peer-reviewed publications (including MICCAI and WACV conference papers), contributing novel methods that improved performance over prior baselines by over 5% in F1-score on benchmark datasets.
- Initiated and drove projects end-to-end; from data collection, data analysis, literature review to experimental design and results interpretation, therefore demonstrating self-motivation, critical thinking, and a strong grasp of AI in healthcare.
- Prepared and presented conference abstracts, presentations, and posters; communicating complex findings clearly.
- Collaborated with researchers at MBZUAI to align AI development with clinical needs, ensuring relevance to real-world healthcare applications.

Data Scientist Intern / Automated Data Capturing

Jul 2023 — Aug 2023

Abu Dhabi Ports Group

Abu Dhabi, UAE

- Spearheaded the full deep learning pipeline; from data acquisition to preprocessing, EDA, model training, testing, and deploying a Streamlit-based proof-of-concept (with an ability to scale), showcasing strong troubleshooting and technical autonomy.
- Cleaned and visualized the datasets, aligning outputs with use cases in areas pertaining to the Logistics Cluster's trucking license plate detection and recognition and automated data capturing of trucks' Mulkiya.
- Achieved over 95% accuracy in license plate recognition using CNN and BiLSTM-based architectures for OCR.
- The developed automated license plate detection and recognition indicated potential gains in gate throughput and operational efficiency by optimizing truck-turn-around time to below 15 minutes.
- Outlined a conceptual system for automated Mulkiya capturing, projecting a reduction in human error from 1–5% to 0.005%.
- Clearly translated complex AI concepts and insights to non-technical stakeholders during presentations, highlighting strong communication skills and the ability to bridge technical and business needs.

Research Assistant Intern / Automated Skeletal Bone Age Assessments

Oct 2018 — Jun 2019

Real Doctor AI Research Centre

Hangzhou, China

- Utilized strong problem-solving and analytical skills to conduct extensive research on Deep Learning techniques for Skeletal Bone Age Assessments, identifying and implementing novel computational methods to support pediatric diagnostic processes.
- Co-authored two peer-reviewed publications on Deep Convolutional Neural Networks for pediatric bone age prediction (birth to age 6), contributing to advancements in medical AI and enhancing prior benchmarks.
- Achieved an average deviation of just 2.75 months between automated and clinical assessments, underscoring the effectiveness and precision of the developed solution.
- Clearly communicated technical findings through both written publications and presentations, reflecting strong skills in conveying complex concepts and engaging with diverse teams.

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Teacher (Science, Math, English)

2013 — 2019

Various Roles

Zimbabwe, South Africa, China

- Developed and delivered tailored educational content to diverse learners across STEM and English language.
- Supervised lab work, assessed student outcomes, and supported learner progression; building a strong foundation in communication, adaptability, and instructional leadership.
- Achieved over 95% verifiable success stories of Chinese students who transitioned from lacking confidence and knowledge to becoming fluent English speakers.
- Achieved 98% pass rate in my Physical Science class in the South African 2014 High School National Senior Certificate Grade 12 Public Examinations.
- Left a lasting impact by inspiring genuine enjoyment for learning and cultivating a positive attitude among my Mathematics students, as evidenced by their testimonials.

STARTUP PROJECT

AI-Powered Financial Documents Table Extraction

2024 — Present

(Ongoing — expanding towards invoices & receipts integration)

Remote

- Identified a market gap in South Africa for automating structured data extraction from complex financial tables in PDFs, validating customer pain points through market research and competitive analysis.
- Designed a scalable, AI-driven architecture combining object detection models, OCR pipelines, and custom heuristics to accurately extract and structure tabular data across diverse financial documents.
- Built a robust end-to-end data pipeline by collecting, labeling, and preprocessing financial reports, enabling model generalization across multiple banking institutions.
- Developed a secure full-stack web application using Node.js, React, Material UI, and REST APIs, enabling real-time client interaction, batch document processing, and enterprise-ready dashboards.
- Planned future expansion to integrate LLM-powered analytics and RAG techniques, enabling intelligent insights, trend analysis, and conversational data querying for invoices, receipts, and financial reports.

SCHOLARSHIPS AND AWARDS

Selected for MBZUAI Full Scholarship with an international acceptance rate of 4.8%

2022

Zhejiang Provincial Government Scholarship Award for Excellent Students

2015, 2016, 2018

Chinese Government Outstanding International Student Scholarship Award

2017

30th Canal Cup Extracurricular Academic Science and Technology Works Competition Award

2018

Silver Prize - 5th Zhejiang Province Internet+ College Student Innovation and Entrepreneurship Competition

2019

VOLUNTARY ACTIVITIES

MBZUAI Graduate Student Council Ambassador

Fall 2023

MBZUAI Environment, Health and Safety (EHS) Committee Member

Fall 2023

ZJUT International Students Hall Assistant

Spring 2016

PUBLICATIONS

- Divergent Domains, Convergent Grading: Enhancing Diabetic Retinopathy Grading Generalization. **Chokuwa S**, Khan MH. 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). IEEE, 2025.
- DGM-DR: Domain Generalization with Mutual Information Regularized Diabetic Retinopathy Classification. A Matsun, D.O Mohamed, **S Chokuwa**, M Ridzuan, M Yaqub. MICCAI Workshop on Domain Adaptation and Representation Transfer, 115-125
- Generalizing Across Domains in Diabetic Retinopathy via Variational Autoencoders. **Chokuwa S**, Khan MH. International Conference on Medical Image Computing and Computer-Assisted Intervention 2023 Oct 8 (pp. 265-274). Cham: Springer Nature Switzerland.
- Skeletal bone age assessment based on deep convolutional neural networks. P Hao, Y Chen, **S Chokuwa**, F Wu, C Bai. Advances in Multimedia Information Processing-PCM 2018: 19th Pacific-Rim Conference on Multimedia, Hefei, China, September 21-22, 2018, Proceedings, Part II 19 Pages 408-417
- Skeletal bone age assessments for young children based on regression convolutional neural networks. P Hao, **S Chokuwa**, X Xie, F Wu, J Wu, C Bai. Mathematical Biosciences and Engineering 16 (6), 6454-6466